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AMERICAN MARKETING ASSOCIATION

*New Technologies in Marketing Special Issue: Article*

Journal of Marketing
2022, Vol. 86(1) 48–66
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DOI: 10.1177/0022242921995449
journals.sagepub.com/home/jmx
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# Augmented Reality in Retail and Its Impact on Sales

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**Abstract**

The rise of augmented reality (AR) technology presents marketers with promising opportunities to engage customers and transform their brand experience. Although firms are keen to invest in AR, research documenting its tangible impact in real-world contexts is sparse. In this article, the authors outline four broad uses of the technology in retail settings. They then focus specifically on the use of AR to facilitate product evaluation prior to purchase and empirically investigate its impact on sales in online retail. Using data obtained from an international cosmetics retailer, they find that AR usage on the retailer's mobile app is associated with higher sales for brands that are less popular, products with narrower appeal, and products that are more expensive. In addition, the effect of AR is stronger for customers who are new to the online channel or product category, suggesting that the sales increase is coming from online channel adoption and category expansion. These findings provide converging evidence that AR is most effective when product-related uncertainty is high, demonstrating the technology's potential to increase sales by reducing uncertainty and instilling purchase confidence. To encourage more impactful research in this area, the authors conclude with a research agenda for AR in marketing.

**Keywords**

augmented reality, mobile app, online retail, product uncertainty, virtual product experience

Online supplement: https://doi.org/10.1177/0022242921995449

> "At some point, we're going to look back and think, how did we not have a digital layer on the physical world?"
>
> – Greg Jones, Director of VR and AR at Google

Augmented reality (AR) is a technology that superimposes virtual objects onto a live view of physical environments, helping users visualize how these objects would fit into their physical world. Even though AR is in its early stages of growth, leaders in the field such as Apple's CEO, Tim Cook, and Google's Director of Virtual Reality (VR) and AR, Greg Jones, have lauded its potential to transform the retail experience (Arthur 2017; Griffin 2017). With the launch of AR toolkits by technology giants Apple and Google, it is now easier for companies to develop their own AR-enabled mobile apps. Jumping on the bandwagon, Facebook recently introduced AR-enabled display advertisements for their News Feed (Carnahan 2019), making the technology even more accessible to companies.

From a retail perspective, a promising application of AR is to facilitate product evaluation by letting customers experience products virtually prior to purchase. Although research has emphasized the importance of direct product experiences to help customers learn about product benefits and assess product fit (Bell, Gallino, and Moreno 2018; Chandukala, Dotson, and Liu 2017), offering direct product experiences can be a logistical challenge, especially in online retail. The introduction of AR has made it possible for shoppers to experience products virtually in the absence of physical products, managing their expectations and instilling purchase confidence (Porter and Heppelmann 2017). For example, Amazon and IKEA are using this technology to help customers determine if products or furniture pieces offered online are compatible with their existing room décor, and L'Oréal and Sephora are using AR to show customers how different cosmetic products would alter their appearance. Some of these applications are illustrated in Web Appendix A.

Despite the keen interest in AR, there has been limited research demonstrating its tangible impact in real-world

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contexts. Understanding the potential for AR to increase revenues is important for justifying investments in this new technology. However, the impact of AR on actual product sales is still ambiguous. By helping customers visualize products in their consumption contexts, AR could reduce product fit uncertainty, resulting in more sales. Conversely, AR may also discourage purchases if it leads to perceptions that the products may not fit well. As the technology is unable to convey experiential product attributes that could be important in purchase decisions (e.g., product texture or scent), the impact of AR on sales could also be insignificant. This uncertainty surrounding the impact of AR has been cited as one of the main reasons why companies are still hesitant to embrace the technology, even though most recognize the exciting opportunities it offers (BCG 2018). Echoing this lack of clarity, a recent CNN article regarding applications of AR in the cosmetics industry expressed that "virtual lipsticks and smokey eye shadows are popular in apps, but are they translating into more makeup sales? Hard data isn't easy to come by" (Metz 2019).

Furthermore, whether and how the impact of AR varies across different products or customer segments is also unclear. Having a more nuanced understanding of how AR affects sales would help marketing managers determine when it would be most appropriate to deploy the technology. Conceivably, if AR increases sales by reducing uncertainty, its impact may depend on product and customer characteristics that influence uncertainty in purchase decisions, such as brand popularity, product appeal, and customers' familiarity with the retail channel or category. Accordingly, the present research adopts the retailers' perspective to examine the following questions:

1. How does the use of AR to facilitate product evaluation impact product sales?
2. How does the sales impact of AR usage differ across product characteristics, such as brand popularity, product appeal, rating, and price?
3. How do customers' prior experiences with the online channel and product category influence the sales impact of AR usage?

Given that AR is predominantly available on mobile apps (Petrock 2020), we focus on the mobile app platform for our analyses. We obtained data from an international cosmetics retailer that incorporated AR into its mobile app to help customers realistically visualize how they would look when they are using different cosmetic products (e.g., eyeshadows, lipsticks). The data contain sales records for 2,300 products, as well as browsing and purchase histories for 160,400 customers, allowing us to investigate how the sales impact of AR varies by product and customer characteristics. In addition, introduction of the AR feature for two product categories during the observation period provided us with a quasi-experimental setting to examine the impact of AR introduction on category sales.

Findings from our research provide preliminary evidence that AR usage has a positive impact on product sales. The overall impact appears to be small, but certain products are more likely to benefit from the technology than others. In particular, the impact of AR is stronger for brands that are less popular and products with narrower appeal, suggesting that AR could level the playing field for niche brands or products (sometimes referred to as products in the "long tail" of the product sales distribution; e.g., Brynjolfsson, Hu, and Simester 2011). The increase in sales is also greater for products that are more expensive, indicating that AR could increase overall revenues for retailers. In addition, customers who are new to the online channel or product category are more likely to purchase after using AR, suggesting that AR has the potential to promote online channel adoption and category expansion. These findings provide converging evidence that AR is most effective when product-related uncertainty is high, implying that uncertainty reduction could be a possible mechanism by which AR could improve sales.

This article is one of the first to empirically demonstrate the impact of AR on sales and how it varies across product and customer characteristics using real-world data. In doing so, it extends prior studies on AR in the marketing field and represents an initial step in understanding what AR means for marketers and retailers. Beyond influencing sales, AR could transform the way brands reach out to and connect with customers at different stages of the customer journey. In the following section, we provide an overview of AR and elaborate on four ways the technology can be incorporated into brands' marketing strategies to reshape the customer retail experience. Then, we focus specifically on how the use of AR to facilitate product evaluation prior to purchase impacts sales in online retail. To encourage marketing academics to further engage in impactful and managerially relevant research in this area, we conclude with a research agenda that we developed in consultation with industry experts and marketing practitioners.

## Augmented Reality

### *Augmented Reality Technology*

Augmented reality integrates virtual elements into real-world environments to create alternate perceptions of reality. Using sensors and object recognition capabilities from input devices such as cameras, AR technology scans the physical environment, identifies features in the environment, and superimposes virtual objects (e.g., two- or three-dimensional images or animations, text, sounds) on top of a live view of the real world. By blending virtual elements into physical environments in real time, AR enriches users' visual and auditory perceptions of reality. In most cases, the virtual elements are also responsive to movements or gestures, creating an interactive experience for users.

Although AR is often classified together with VR, the two technologies are distinct, both in how they function and the way they are experienced. Unlike AR, which receives input from the real world and adds virtual elements to it, VR immerses users in a completely digital and artificial environment, shutting them out from their surroundings. Due to the

**Table 1.** Uses of AR in Retail.

| Uses of AR | Role of AR | Illustrative Use Cases |
|---|---|---|
| Entertain customers | • Create novel and engaging experiences for customers<br>• Build brand interest<br>• Drive foot traffic to physical stores | • Walmart collaborated with DC Comics and Marvel to bring exclusive superhero-themed AR experiences to selected outlets.<br>• Starbucks Reserve Roastery in Shanghai uses AR to offer customers a digital tour of their massive roasting facility. |
| Educate customers | • Deliver content and information in an interactive and visually appealing manner<br>• Help customers understand complex mechanisms and better appreciate the value of products | • Walgreen's and Lowe's use AR in their in-store navigation apps to guide users to product locations and notify them if there are special promotions along the way.<br>• Toyota and Hyundai use AR to demonstrate key features and innovative technologies in their new car models. |
| Help customers evaluate product fit | • Help customers visualize products in their actual consumption contexts<br>• Increase customers' confidence in their purchase decisions in the absence of physical products<br>• Accommodate wide product assortments and customization without the need for physical inventory | • IKEA's Place app uses AR to help customers determine if products fit with their existing room décor.<br>• L'Oréal's Virtual Try-On feature and Sephora's Virtual Artist app use AR to show customers how different cosmetic products would look on them.<br>• Uniqlo and Topshop use AR to offer a more convenient way of trying on different outfits in their physical stores.<br>• BMW and Audi use AR to give customers a preview of cars based on customizable features such as paint color, wheel design, and interior aesthetics. |
| Enhance customers' postpurchase consumption experience | • Offer new ways of enjoying products after they are purchased<br>• Deliver additional information while the products are being used or consumed | • LEGO's Hidden Side sets are specially designed to be played together with the companion AR app.<br>• McDonald's used AR to let customers discover the origins of ingredients in the food they purchased.<br>• Hyundai's Virtual Guide app uses AR to teach car owners how to perform basic maintenance. |

*Note*: URL links to these examples are provided in Web Appendix B.

disorienting experience of being entirely isolated from the real world and the expensive headsets required (Ericsson 2017), the appeal of VR has largely been limited to industries with products high in simulated content, such as gaming and entertainment (Chaykowski 2018). In contrast, AR allows users to experience virtual elements without the vulnerability of being blind to the real world. In addition, AR can be experienced directly from handheld devices that users already own (e.g., tablets or smartphones). Thus, AR is rapidly gaining prominence, and close to 100 million U.S. consumers are expected to use the technology regularly by 2022 (Petrock 2020).

## Augmented Reality in Retail

The unique capabilities of AR present marketers with new opportunities to engage customers and transform the brand experience. Drawing on an extensive review of current applications of AR, we identified four broad uses of the technology in retail settings: to (1) entertain and (2) educate customers, help them (3) evaluate product fit, and (4) enhance the postpurchase consumption experience. These uses loosely correspond to customers' journey from awareness to interest, consideration, purchase, and consumption, and they may not be mutually exclusive. Next, we elaborate on these four uses and provide a summary with relevant examples in Table 1.[1]

*Entertain.* AR's ability to transform static objects into interactive and animated three-dimensional objects offers new ways for marketers to create fresh experiences to captivate and entertain customers. Besides generating hype and interest, marketers have also used AR-enabled experiences to drive traffic to their physical locations. For example, Walmart collaborated with media companies such as DC Comics and Marvel to bring exclusive superhero-themed AR experiences to their stores by placing special thematic displays in selected outlets. In addition to creating novel and engaging experiences for customers, it also encouraged them to explore different areas within the stores.

*Educate.* Due to its interactive and immersive format, AR is also an effective medium for delivering content and information to customers. For instance, to help customers better appreciate their new car models, Toyota and Hyundai have utilized AR to demonstrate key features and innovative technologies in a vivid and visually appealing manner. Retailers can also use AR

[1] URL links to these examples are provided in Web Appendix B.

to help customers navigate stores or highlight relevant product information to influence their in-store purchase decisions. Companies such as Walgreen's and Lowe's have developed in-store navigation apps that overlay directional signals onto a live view of the path in front of users to guide them to product locations and notify them if there are special promotions along the way.

*Evaluate.* By retaining the physical environment as a backdrop to virtual elements, AR also helps users visualize how products would appear in their actual consumption contexts, allowing them to more accurately assess product fit prior to purchase. For example, IKEA's Place app uses AR to give customers a preview of different furniture pieces in their homes by overlaying true-to-scale, three-dimensional models of products onto a live view of the room. Customers can easily determine if a product fits in a given space without the hassle of taking measurements. Fashion retailers Uniqlo and Topshop have also deployed the same technology in their physical stores, offering customers greater convenience by reducing the need for them to change in and out of different outfits. An added advantage of AR is its ability to accommodate a wide assortment of products. By replacing tangible product displays with lifelike virtual previews of products, retailers can overcome the constraints of physical space while still offering customers the opportunity to explore different product options. This capability is particularly useful for made-to-order or bulky products. Car manufacturers BMW and Audi have used AR to provide customers with true-to-scale, three-dimensional visual representations of car models based on customizable features such as paint color, wheel design, and interior aesthetics. These cases exemplify AR's huge potential to increase customers' confidence in their purchase decisions for a variety of products.

*Enhance.* Lastly, AR can be used to enhance and redefine the way products are experienced or consumed after they have been purchased. For example, LEGO recently launched several brick sets that are specially designed to combine physical and virtual gameplay. Through the companion AR app, animated LEGO characters spring to life and interact with the physical LEGO sets, creating a whole new playing experience. In a bid to address skepticism about the quality of its food ingredients, McDonald's has also used AR to let customers discover the origins of ingredients in the food they purchased via storytelling and three-dimensional animations.

The present research focuses on the use of AR to help customers evaluate products prior to purchase. Specifically, we explore the possibility of leveraging AR to reduce product-related uncertainty in online purchase decisions. To extend prior research on AR in retail (summarized in Table 2), we use real-world data to examine how customers' use of AR to try products (for brevity, we refer to this as "AR usage" for the rest of the article) affects product and brand sales. In the following section, we present our conceptual framework and develop hypotheses for the impact of AR usage on sales.

## Conceptual Framework

### *Product Uncertainty in Online Retail*

Because customers cannot perfectly predict the consequences of their purchase decisions, uncertainty is inherent in market exchanges (Bauer 1960). However, it is especially pronounced in online environments due to the spatial separation between buyers and sellers as well as the temporal separation between payment and product fulfillment (Burke 2002; Pavlou, Liang, and Xue 2007). Unlike in traditional retail settings, customers are unable to physically inspect or evaluate products before making a purchase, resulting in greater uncertainty that the products would be able to deliver the expected level of performance or benefits (Bell, Gallino, and Moreno 2018; Dimoka, Hong, and Pavlou 2012; Kim and Krishnan 2015).

Researchers have broadly distinguished between two types of product uncertainty in online markets: product performance uncertainty and product fit uncertainty. Product performance uncertainty occurs when customers are unable to evaluate or predict product performance due to imperfect knowledge (Dimoka, Hong, and Pavlou 2012). In contrast, product fit uncertainty occurs when customers are unable to determine if the product matches their needs (Bell, Gallino, and Moreno 2018; Hong and Pavlou 2014). The latter form of uncertainty is typically higher for products with experience attributes (i.e., attributes that can only be evaluated after the product has been experienced; Hong and Pavlou 2014), such as apparel or beauty products.

Several mechanisms to reduce product performance uncertainty in online retail have been suggested. For example, retailers could lower information asymmetry by providing diagnostic product descriptions or by including credibility signals such as third-party product assurances, warranties, or customer reviews (Dimoka, Hong, and Pavlou 2012; Weathers, Sharma, and Wood 2007). In contrast, product fit uncertainty typically requires direct product experience to resolve, as it is idiosyncratic in nature and varies from individual to individual. Although some retailers have adopted try-before-you-buy programs (e.g., Warby Parker's home try-on program; Bell, Gallino, and Moreno 2018) or lenient product return policies (Gu and Tayi 2015; Wood 2001) to provide opportunities for direct product experiences, these measures are notoriously costly for retailers due to the additional shipping and handling costs and risks of product damage (Gray 2019). Furthermore, direct product experiences may not be viable or appropriate for certain products, such as products that are customized (e.g., engagement rings), products that require assembly (e.g., furniture), or personal care products (e.g., cosmetics).

### *Augmented Reality and Product Uncertainty*

The introduction of AR has made it possible to substitute direct product experiences with virtual product experiences to facilitate product evaluation and reduce product fit uncertainty. Using a situated cognition perspective, Hilken et al. (2017) propose that the value of AR lies in its ability to help customers

**Table 2.** Selected Literature on AR in Retail.

| Article | Methodology | Context | Key Outcome Variables | Key Findings |
|---|---|---|---|---|
| Hilken et al. (2017) | Experimental | Using situated cognition theory to understand AR's potential to enhance online experiences | Value perceptions of online experiences; decision comfort; purchase and word-of-mouth intentions | • The combination of simulated physical control and environmental embedding offered by AR creates a feeling of spatial presence.<br>• As a result, AR enhances consumers' perceptions of online experiences, decision comfort, and behavioral intentions. |
| Yim, Chu, and Sauer (2017) | Experimental | Comparing AR versus web-based product presentations | Attitude toward AR and purchase intentions | • Compared to web-based displays, AR is more immersive due to its interactive and vivid nature.<br>• As a result, AR is perceived to be more useful and enjoyable, leading to positive consumer attitudes and purchase intentions. |
| Brengman, Willems, and Van Kerrebroeck (2019) | Experimental | Impact of AR on perceived ownership | Perceived ownership and purchase intentions | • Compared to other touch and nontouch interfaces, mobile-enabled AR creates higher feelings of perceived ownership, positively affecting consumers' attitudes and purchase intentions. |
| Heller et al. (2019a) | Experimental | Using mental imagery theory to understand how AR influences word of mouth | Word-of-mouth intentions | • AR improves processing fluency by facilitating imagery generation and transformation, leading to higher consumer decision comfort and word-of-mouth intentions. |
| Heller et al. (2019b) | Experimental | Comparing touch versus voice control modalities in multisensory AR | Decision comfort and willingness to pay | • Touch control (vs. voice control) reduces mental intangibility, leading to higher consumer decision comfort and willingness to pay. |
| Hilken et al. (2020) | Experimental | Shared decision making using social AR | Decision makers' product choice; spillover effects to recommenders | • AR empowers recommenders by allowing them to take the point of view of the decision makers.<br>• AR stimulates recommenders' desire for products, leading to positive behavioral intentions. |
| Current article | Instrumental variable estimation and quasi-experiment using real-world data | Examining the impact of AR usage on sales and the moderating impact of product and customer characteristics | Product and category sales | • AR has a positive impact on sales for brands that are less popular, products with narrower appeal, and products that are more expensive.<br>• AR has a stronger impact for customers who are new to the retailer's online channel or product category. |

visually integrate virtual products into the real-world environment (i.e., "environmental embedding") and use bodily movements and physical actions to control how products are presented (i.e., "simulated physical control"). The unique combination of these two properties induces perceptions that the virtual products are physically present in the real world, creating realistic product experiences. Consequently, customers are able to evaluate products as if they are actually interacting with the real products, resulting in reduced product fit uncertainty. In line with this, prior research finds that vivid images and greater control over the presentation of information are effective ways to alleviate uncertainty in online environments (Weathers, Sharma, and Wood 2007). By helping customers visualize products in their consumption contexts and reducing product fit uncertainty, AR-enabled product experiences increase the level of ease customers feel in the decision-making process, translating to positive behavioral intentions (Heller et al. 2019a; Hilken et al. 2017).

However, although AR communicates visual information about products, it is unable to convey other experiential product attributes (e.g., product texture, scent). For example, even though customers may use AR to visualize an IKEA sofa in a room, they are unable to assess how comfortable it is. Similarly, users trying on cosmetic products via AR are unable to evaluate other product attributes such as the texture and consistency of the product, which may affect ease of application and the way the product feels on the skin. According to Kempf and Smith (1998), if customers do not perceive trial experiences as accurately representing actual consumption experiences, they may discount those trial experiences when they form judgments about the product. Thus, the extent to which virtual product experiences involving AR could influence online purchases is unclear. Nevertheless, as prior research has demonstrated the positive effects of providing fit information in online retail (e.g., Gallino and Moreno 2018; Kim and Forsythe 2008), we expect AR usage to have a positive impact on product sales because the technology could convey visual information that may reduce product fit uncertainty in online purchase decisions. Therefore, we predict the following:

**$H_1$:** AR usage has a positive impact on sales.

Building on the proposition that AR usage increases sales by reducing product fit uncertainty, we further hypothesize that AR would have a stronger impact when customers experience higher levels of uncertainty. In particular, the level of uncertainty experienced in a purchase decision could depend on product characteristics such as brand popularity, product appeal, and ratings. The level of uncertainty may also influence the price that customers are willing to pay for the product. Thus, the relationship between AR usage and sales may differ across these product characteristics. In addition, customers also vary in their need to reduce product fit uncertainty before making a purchase (Bell, Gallino, and Moreno 2018). This need to reduce uncertainty could depend on customers' familiarity with the online channel and product category. As a result, the impact of AR may also vary across these customer characteristics. Accordingly, we develop hypotheses for the moderating effects of product and customer characteristics in the following sections. Our conceptual framework is presented in Figure 1.

## Moderating Effects of Product Characteristics

*Brand popularity.* Prior research has shown that consumers are more cautious when they purchase from lesser-known brands, as they anticipate feeling more regret if the product turns out to be inferior (Simonson 1992). Consistent with this, Erdem, Swait, and Valenzuela (2006) find that cultures high in uncertainty avoidance place greater emphasis on brand credibility. In online environments, brand signals are even more important because consumers are not able to inspect products before purchasing (Danaher, Wilson, and Davis 2003). However, Hollenbeck (2018) demonstrates that when additional information is available to facilitate decision making, consumers rely less on brand signals. As a result, less-established brands benefit more from the increased availability of information. In the same vein, by communicating visual information to help customers assess product fit, AR may reduce uncertainty in online purchase decisions. Consequently, AR may decrease customers' reliance on brand signals and inadvertently increase preference for brands that are less popular. We use the term "popular" in a general sense to refer to brands that are more widely adopted. Therefore, we hypothesize the following:

**$H_{2a}$:** The impact of AR usage on sales is stronger for brands that are less popular.

*Product appeal.* Within the same category or brand, products may also have different levels of appeal due to the alignment between their inherent characteristics and general consumer preferences. For example, a red lipstick is more mainstream and has broader appeal than a blue lipstick. We draw a distinction between brand popularity and product appeal in that the latter depends on intrinsic properties of the product and could be independent of the brand. Thus, a red lipstick from an unknown brand could have broad appeal but low brand popularity, whereas a blue lipstick from a well-known brand could have limited appeal despite having high brand popularity. As products with broad appeal cater to the masses, they are more likely to match the needs of the general consumer. Conversely, because products with narrower appeal serve a niche segment, there is a higher probability that they will not match the preferences of the general consumer and will thus carry greater product fit uncertainty. Nevertheless, Brynjolfsson, Hu, and Simester (2011) demonstrate that in online contexts, search and discovery features such as search tools or recommendation engines can shift consumers' preferences to niche products by lowering the cost of acquiring product information. Consistent with this, Tucker and Zhang (2011) find that products with narrower appeal benefit more from greater information

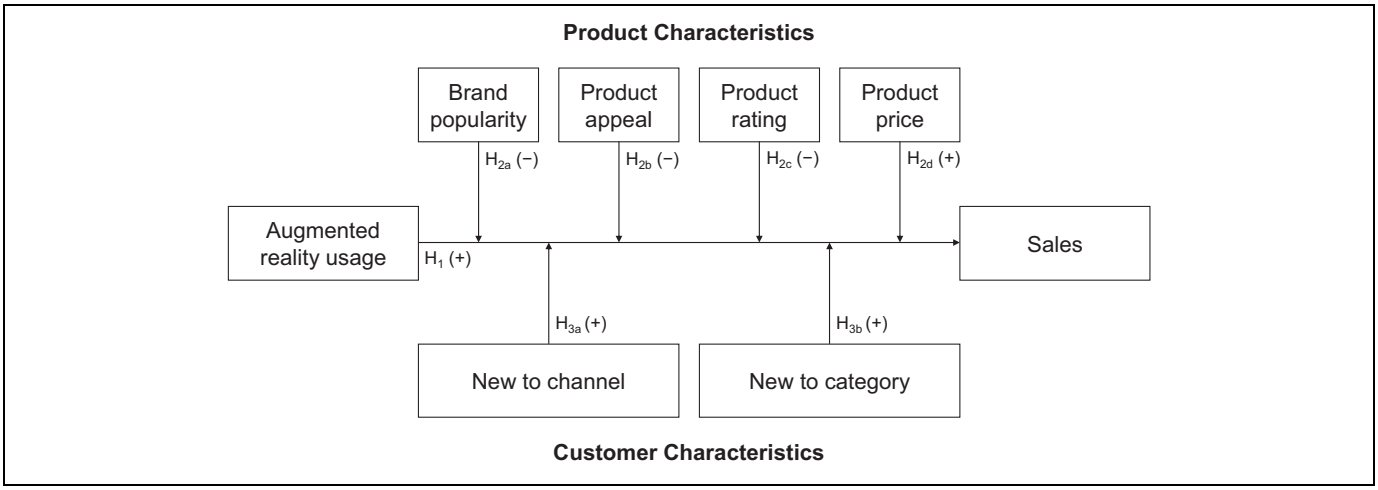



**Figure 1.** Conceptual framework.
*Note*: Signs in parentheses represent the hypothesized effects.

availability. By visually conveying product information to help customers assess product fit in an effortless and risk-free environment, AR could have a stronger impact for products with narrower appeal due to the higher product fit uncertainty associated with these products. Therefore, we hypothesize the following:

> **$H_{2b}$:** The impact of AR usage on sales is stronger for products with narrower appeal.

*Ratings.* Customers often turn to online ratings or reviews as a source of information to resolve uncertainty about product quality and fit (Chen and Xie 2008). In line with this, Kübler et al. (2018) find that consumers from countries that are high in uncertainty avoidance are more sensitive to both the valence and volume of product ratings. However, as consumers tend to overrate direct experiences with products (Hoch 2002), the ability to evaluate products and resolve uncertainty via first-hand experiences with those products on AR platforms may reduce customers' reliance on online ratings. Thus, by enabling customers to learn about product benefits and assess product fit through their own virtual experiences, AR could diminish the role of online ratings in purchase decisions. As a result, customers may be more amenable to purchasing products despite their lower ratings if they are able to try these products using AR. Therefore, we predict the following:

> **$H_{2c}$:** The impact of AR usage on sales is stronger for products with lower ratings.

*Price.* When customers experience product uncertainty, they are not able to accurately assess the benefits the products offer. As a result, customers are more likely to undervalue the products and are less willing to pay a premium (Dimoka, Hong, and Pavlou 2012). Consistent with this, Kim and Krishnan (2015) find that customers who are familiar with online shopping are still hesitant to purchase expensive products through the internet when there is a high degree of product uncertainty because they could suffer greater financial losses if these products do not fit them well. By facilitating product evaluation prior to purchase, AR helps customers ascertain if products match their needs and preferences. Consequently, customers may experience less uncertainty and feel more comfortable purchasing products that are more expensive. In line with this, Heller et al. (2019b) find that AR usage improves decision comfort, leading to higher willingness to pay. Therefore, we predict the following:

> **$H_{2d}$:** The impact of AR usage on sales is stronger for more expensive products.

## *Moderating Effects of Customer Characteristics*

*Channel experience.* According to Kim and Krishnan (2015), customers who are familiar with online shopping are more inclined to purchase products with a higher degree of uncertainty because their cumulative online shopping experiences help them develop the ability to assess products when limited information is available. Thus, customers who have purchased from a retailer's online channel in the past may feel more comfortable making subsequent online purchases despite experiencing product uncertainty, potentially making them less dependent on AR to make their purchase decisions. In contrast, customers who are new to the retailer's online channel (but have made prior purchases at the retailer's offline channel) are not accustomed to making purchases in the absence of actual products. As a result, they may experience greater product fit uncertainty and may be deterred from purchasing online due to the inability to assess product fit. Because AR simulates the in-store experience of trying products, it may help reduce product fit uncertainty for customers who are new to the online channel. These customers may derive greater value from the ability to evaluate products virtually, potentially

making them more likely to purchase online after using AR. Therefore, we predict the following:

> **$H_{3a}$:** The impact of AR usage on sales is stronger for customers who are new to the retailer's online channel.

*Category experience.* Besides channel experience, customers' familiarity with the product category also affects their level of product fit uncertainty (Hong and Pavlou 2014). Customers who are familiar with a product category can draw on their prior experiences as an information source to form judgments about products (Smith and Swinyard 1982). As a result, they may rely less on AR in their purchase decisions. Conversely, customers who are unfamiliar with a product category lack the necessary category knowledge to evaluate product attributes and, at the same time, may not be aware of their own preferences (Hong and Pavlou 2014). Consequently, these customers will have more difficulty assessing whether a product's attributes match their preferences, resulting in greater product fit uncertainty. By helping customers visualize how products would appear in their actual consumption contexts, AR could reduce product fit uncertainty and increase purchase confidence for customers who are new to the product category. As a result, AR usage may have a stronger impact on the purchase decisions for these customers. Therefore, we predict the following:

> **$H_{3b}$:** The impact of AR usage on sales is stronger for customers who are new to the product category.

To summarize, we propose that AR usage will positively impact sales by reducing product uncertainty. Following this line of reasoning, we developed several predictions about which products would be more likely to benefit from AR and which customers would be more likely to respond to AR.

## Empirical Analysis

### Empirical Context

As AR is predominantly available through mobile apps (Petrock 2020), we focus our analyses on the mobile app platform. To test our hypotheses, we obtained data from an international cosmetics retailer with both an online and offline presence. Leveraging AR technology, the retailer integrated a new feature on its existing mobile app that allowed customers to virtually try on makeup products (e.g., eyeshadows, lipsticks). The AR technology detected customers' facial features via their smartphone cameras and superimposed the shade of chosen products onto a live view of their face in real time, giving them a realistic visual representation of their appearance when they are using the products. The brand, product name, and price were displayed at the top of the screen. Figure A3 in Web Appendix A provides a visual example of a customer trying on a lipstick using the AR feature. For comparison, the corresponding product detail page view (i.e., the conventional way of conveying product-related information on mobile retail apps) is also provided. Prior to the start of our observation period in December 2017, the AR feature was only available for lip categories (i.e., lipstick and lip gloss) and was subsequently introduced for eye categories (i.e., eyeshadow and eyeliner) in March 2018. Figure A4 in Web Appendix A provides a visual overview of AR availability for the different categories.

We obtained two separate data sets from the retailer for one of its key markets in Asia Pacific. The first data set contained information about browsing activities on the mobile app, including specific products customers tried using the AR feature, and covered a 19-month period from December 2017 to June 2019. The second data set contained transaction records from June 2017 to June 2019 for all retail channels, including mobile app, website, and offline stores. We merged the two using customers' loyalty card number, which allowed us to match AR usage and product purchases at a disaggregate level.

During the 19-month period, a total of 160,407 shoppers browsed products from the lip and eye categories across 806,029 sessions, 20.8% of which involved AR usage. Customers who used AR during the session spent 20.7% more time browsing ($M_{with_AR} = 16.6$ minutes, $M_{without_AR} = 13.8$ minutes, $p < .01$) and browsed 1.28 times more products ($M_{with_AR} = 53.9$, $M_{without_AR} = 42.2$, $p < .01$). The purchase rate for sessions with AR usage was 19.8% higher than for sessions without AR usage (3.15% with AR vs. 2.63% without AR, $p < .01$), providing preliminary indication of the positive impact of AR on sales.

We divide our analyses into three sections. In the first section, we perform the analysis at the product level to examine the moderating effects of brand popularity, product appeal, rating, and price. To minimize selection bias arising from availability of the AR feature, we focus on lipsticks and lip glosses, as the feature was available for $> 96\%$ of products in each of these categories. In the second section, we take advantage of the introduction of AR for two eye categories (i.e., eyeshadow and eyeliner) to examine the effect of AR introduction on category sales using a quasi-experimental differences-in-differences-in-differences (DDD) approach. Finally, we investigate how the impact of AR varies at the customer level. As all customers had no knowledge that the AR feature would be introduced for the two eye categories prior to the introduction, the event provided us with a clean setting for examining how customers' channel and category experience (prior to the introduction) would moderate the impact of AR usage on purchase probability.

### Product-Level Analysis

As product color is an important factor in cosmetic purchases, we considered each shade/color of retail merchandise as a unique product. In total, we had 2,334 products in the lipstick and lip gloss categories (1,984 products across 41 brands for lipstick; 350 products across 28 brands for lip gloss). Our empirical strategy was to relate the number of customers using AR to try each product during a particular time period with sales volume for that product during the same time period. We estimated the model at the monthly product level, giving us a

**Table 3.** Variable Operationalization and Descriptive Statistics for Product Model.

| Variable | Operationalization | Mean | SD | Min | Median | Max |
|---|---|---|---|---|---|---|
| Product Sales | Total product sales from mobile app (in units) | .46 | 1.73 | .00 | .00 | 64.00 |
| AR Usage | Number of customers using AR to try the product in focal country | 13.90 | 22.67 | .00 | 6.00 | 611.00 |
| AR Usage$^{Alt}$ | Number of times the product was tried using AR in focal country | 14.45 | 23.92 | .00 | 7.00 | 620.00 |
| Brand Popularity | Total brand sales from website (in thousands of units) | .04 | .07 | .00 | .02 | .51 |
| Brand Popularity$^{Alt}$ | Number of customers buying the brand from website (in thousands) | .03 | .04 | .00 | .01 | .42 |
| Appeal | Total product sales from website (in units) | .25 | .97 | .00 | .00 | 32.00 |
| Appeal$^{Alt}$ | Number of customers buying the product from website | .25 | .95 | .00 | .00 | 31.00 |
| Rating | Product rating at time t (on a five-point scale) | 4.14 | .67 | .50 | 4.25 | 5.00 |
| Price | Product price at time t | 31.80 | 11.17 | 5.00 | 32.00 | 77.00 |
| AR Usage$^{Country_A}$ | Number of customers using AR to try the product in Country A | .59 | 1.43 | .00 | .00 | 39.00 |
| AR Usage$^{Country_B}$ | Number of customers using AR to try the product in Country B | .28 | .71 | .00 | .00 | 11.00 |

total of 44,346 observations (2,334 products × 19 months from December 2017 to June 2019). As one of our objectives was to examine the moderating effect of product ratings, we included products with a rating in the main analysis and replicated the analysis for all products as a robustness check. Our final sample for the main analysis consisted of 29,345 observations.

*Model specification.* For each product i, we modeled how the volume of AR usage in month t, AR $\text{Usage}_{it}$, influenced the number of products sold in month t, Product $\text{Sales}_{it}$. As Product $\text{Sales}_{it}$ was a count variable with significant over-dispersion (M = .46, SD = 1.73) and over 80% of observations were 0, we used a zero-inflated negative binomial model for the estimation. The vector of covariates in the regression is given by the following equation:

$$\begin{aligned}\mathbf{X_{it}}\beta = {} & \beta_0 + \beta_1 \text{AR Usage}_{it} + \beta_2 \text{Brand Popularity}_{it} \\ & + \beta_3 \text{Appeal}_{it} + \beta_4 \text{Rating}_{it} + \beta_5 \text{Price}_{it} \\ & + \beta_6 \text{AR Usage}_{it} \times \text{Brand Popularity}_{it} \\ & + \beta_7 \text{AR Usage}_{it} \times \text{Appeal}_{it} \\ & + \beta_8 \text{AR Usage}_{it} \times \text{Rating}_{it} \\ & + \beta_9 \text{AR Usage}_{it} \times \text{Price}_{it} \\ & + \beta_{10} \text{Category}_i + \sum_{m=1}^{T-1} \delta_m \text{Month}_t + \varepsilon_{it}. \end{aligned} \quad (1)$$

In Equation 1, we measured AR $\text{Usage}_{it}$ as the number of customers using AR to try product i during month t. As brands that are more widely adopted should have higher sales, and because the web and app channels are both online and carry identical products, we used total brand sales (within the category) from the web channel during the same period as a proxy for brand popularity, Brand $\text{Popularity}_{it}$. Following prior research using product sales as an indicator of mass or niche appeal (e.g., Brynjolfsson, Hu, and Simester 2011), we used total product sales from the web channel during the same period to reflect product i's breadth of appeal, $\text{Appeal}_{it}$. $\text{Rating}_{it}$ and $\text{Price}_{it}$ were the rating and price of product i at time t, respectively. To examine how the impact of AR was influenced by brand popularity, product appeal, rating, and price, we included the corresponding interactions in the model. In addition, we included $\text{Category}_i$ (1 = lipstick, 0 = lip gloss) and a series of dummy variables, $\text{Month}_t$, (for t = 1, . . . , T months) to control for category and month effects. Table 3 provides a summary of how the variables were operationalized and their descriptive statistics, and we provide their correlations in Web Appendix C. All the correlations were low, and the variance inflation factors were below 1.62, indicating that multicollinearity was not an issue. To prevent overestimation of effects due to the panel structure of the data, we clustered standard errors at the product level (e.g., Tucker 2014).

*Identification strategy.* Our objective was to understand how the volume of AR usage for product i during month t, AR $\text{Usage}_{it}$, influenced product sales, Product $\text{Sales}_{it}$. However, AR $\text{Usage}_{it}$ could be endogenous, as customers may have been more inclined to use AR to try products they were already interested in purchasing. To account for this endogeneity, we used the two-stage residual inclusion method (Terza, Basu, and Rathouz 2008), which has been used in recent research when both the endogenous and dependent variables are nonlinear (e.g., Arora, ter Hofstede, and Mahajan 2017; Danaher et al. 2020).

Following the two-stage residual inclusion method, we first regressed the endogenous variable, AR $\text{Usage}_{it}$, on all other covariates in Equation 1. Residuals from this first stage were then included to estimate Product $\text{Sales}_{it}$. Similar to the control function approach (Petrin and Train 2010), the included residuals controlled for the portion of the endogenous variable that would otherwise correlate with the error term in Equation 1. According to Terza, Basu, and Rathouz (2008), we needed to include instruments in the first stage estimation to resolve the identification problem in Equation 1. These instruments should (1) be strongly related to the endogenous variable and (2) not be correlated with the error term in Equation 1. In other words, the instruments should only have an indirect relationship with the outcome variable, Product $\text{Sales}_{it}$, through their association with the endogenous variable, AR $\text{Usage}_{it}$. As realizations of the same variable from different markets can serve as suitable instruments (Papies, Ebbes, and Van Heerde 2017, p. 601), we used the volume of AR usage in two other countries for the

**Table 4.** Product Model: Impact of AR Usage on Product Sales and Moderating Effects of Product Characteristics.

| | Column 1 Second Stage (Without Interactions) | Column 2 Second Stage (Full Model) | Column 3 Alternative Instrument for AR Usage | Column 4 Including Products Without Ratings |
|---|---|---|---|---|
| **AR Usage ($H_1$)** | **.006 (.001) ***** | −.002 (.006) | −.003 (.007) | .007 (.006) |
| Brand Popularity | .894 (.364) ** | 1.482 (.356) *** | 1.796 (.396) *** | 1.675 (.333) *** |
| Appeal | .385 (.029) *** | .416 (.023) *** | .419 (.027) *** | .473 (.023) *** |
| Rating | .094 (.042) ** | .052 (.054) | .062 (.056) | – |
| Price | −.005 (.003) * | **−.009 (.004) **** | **−.008 (.004) **** | **−.010 (.003) ***** |
| **AR Usage × Brand Popularity ($H_{2a}$)** | – | **−.022 (.009) **** | **−.025 (.008) ***** | **−.028 (.010) ***** |
| **AR Usage × Appeal ($H_{2b}$)** | – | **−.001 (.000) ***** | **−.001 (.000) **** | **−.001 (.000) ***** |
| **AR Usage × Rating ($H_{2c}$)** | – | **.001 (.001)** | **.001 (.001)** | **–** |
| **AR Usage × Price ($H_{2d}$)** | – | .000 (.000) * | .000 (.000) * | .000 (.000) * |
| Correction term | .071 (.027) *** | .050 (.027) * | .102 (.053) * | .063 (.026) ** |
| Constant | −1.114 (.223) *** | −.913 (.264) *** | −1.185 (.280) *** | −.687 (.162) *** |
| Category dummy | Included | Included | Included | Included |
| Month dummies | Included | Included | Included | Included |
| Observations | 29,345 | 29,345 | 28,305 | 44,346 |
| Log likelihood | −18,573 | −18,517 | −17,630 | −25,310 |

*$p \leq .10$; **$p \leq .05$; ***$p \leq .01$.
*Note*: Standard errors (clustered at product level) are in parentheses.

same product during the same month as our instruments (i.e., AR $\text{Usage}_{it}^{\text{Country_A}}$ and AR $\text{Usage}_{it}^{\text{Country_B}}$, respectively). Underlying this choice of instruments is the assumption that customer preferences are similar across markets and that product-specific factors affecting customers' interest in trying products using the AR feature should be constant in all markets, satisfying the first condition. However, the number of customers using the AR feature to try products in other markets should have no bearing on customers' purchase decisions in the focal market, satisfying the second requirement. We also used lagged values of AR $\text{Usage}_{it}$ as an alternative instrument (e.g., Danaher et al. 2020) and discuss this further in the robustness analyses section.

Because AR $\text{Usage}_{it}$ is a count variable with significant over-dispersion (M = 13.9, SD = 22.7), we used a negative binomial model for the first stage estimation. As predicted residuals from the first stage were used in the estimation of Equation 1, standard errors needed to be corrected to account for this additional source of variation (Petrin and Train 2010). We implemented the cluster bootstrapping method (Cameron and Miller 2015, p.327) to approximate the correct standard errors using 1000 bootstrap samples.

*Results.* From the first stage estimation (provided in Web Appendix D), coefficients for the instruments are positive and significant (.414 for AR $\text{Usage}^{\text{Country_A}}$ and .301 for AR $\text{Usage}^{\text{Country_B}}$, $p < .01$ for both). Furthermore, the instruments are highly correlated with AR Usage (.75 for AR $\text{Usage}^{\text{Country_A}}$ and .64 for AR $\text{Usage}^{\text{Country_B}}$, $p < .01$ for both), and the F-statistic of excluded instruments in the first stage regression is 5,520, which is well above the recommended cutoff of 10 (Angrist and Pischke 2009). These results indicate that the instruments are strongly related with the endogenous variable. To assess validity of the instruments, we performed the Hansen J-test for over-identifying restrictions. Results from the test fail to reject the null hypothesis that the instruments are uncorrelated with the second stage error term ($\chi^2$ (1) = .699, $p = .40$), providing additional support for the choice of instruments.

To examine the main effect of AR usage in $H_1$, we estimated the second stage model without interaction terms. Results for this model are presented in Table 4, Column 1. The coefficient for AR Usage is significantly positive (.006, $p < .01$), suggesting a small but positive relationship between the number of customers using AR to try the product and sales for that product during the same month. Thus, $H_1$ is supported. The coefficients for other variables are largely in line with common intuition. For example, brand popularity (.894, $p < .05$), breadth of product appeal (.385, $p < .01$), and product rating (.094, $p < .05$) are positively associated with product sales, whereas price (−.005, $p < .10$) has a negative relationship with product sales. The coefficient for the residual correction term, which is equivalent to the Hausman test for the presence of endogeneity (Papies, Ebbes, and Van Heerde 2017), is significant (.071, $p < .01$), indicating that the endogeneity-corrected estimates are preferred. Thus, we focus on results from the two-stage model and provide results for the uncorrected model in Web Appendix D.

Results for the full second stage model are presented in Table 4, Column 2. In support of $H_{2a}$ and $H_{2b}$, the interactions between AR Usage and Brand Popularity (−.022, $p < .05$) and Appeal (−.001, $p < .01$) are significantly negative, indicating that the sales impact of AR usage is stronger for brands that are less popular and products with narrower appeal. The interaction between AR Usage and Price is significantly positive (.000, $p < .10$), suggesting that the sales impact of AR usage is

stronger for products that are more expensive. Thus, $H_{2d}$ is also supported. However, the results do not provide support for $H_{2c}$, as the interaction between AR Usage and Rating is not significant (.001, $p > .10$).

*Robustness analyses.* We performed several analyses to ensure that our findings are robust to different assumptions and model specifications. First, following prior research, which has used lagged values of endogenous variables as instruments (e.g., Danaher et al. 2020), we used the volume of AR usage for product i in the past one month as an alternative instrument. As app activity data prior to the first month (i.e., December 2017) was unavailable, we excluded observations for the first month. Results for this model are presented in Table 4, Column 3, and the findings are consistent. Because we were interested in the moderating effect of ratings, we focused on products that had a rating in the main analysis. Since the coefficient for Rating was not significant, we excluded it in the model specification and replicated the analysis for all products. Results for this model are also consistent with the main findings and are presented in Table 4, Column 4.

We also explored alternative operationalizations for AR Usage, Brand Popularity, and Appeal. Instead of operationalizing AR Usage as the number of customers using AR to try product i, we used the number of times product i was tried using AR to account for repeated AR usage from the same customer. We also operationalized Brand Popularity and Appeal as the number of customers purchasing the brand and product, respectively. Results for these models are reported in Web Appendix E. Across all robustness analyses, results are generally consistent with the main model, providing further validation for our findings.

## Category-Level DDD Analysis

The introduction of AR for two eye categories (i.e., eyeshadow and eyeliner) in mid-March 2018 presented a unique opportunity to examine the impact of AR introduction on sales. Using a quasi-experimental approach, we regarded AR introduction as a treatment and examined its impact by comparing differences in sales for products with and without the AR feature, before and after the feature was introduced. Because the AR feature was only available for eyeshadows and eyeliners, a potential comparison could be between these categories and other eye categories that did not have the feature (i.e., eyebrows, mascaras, and eye palettes). This between-category comparison relies on the crucial assumption that sales trends across different eye categories would be parallel in the absence of AR introduction. As cosmetic products are often used concurrently, sales for products targeting the same facial feature should generally move in the same direction. Since the AR feature was only available on the mobile app, an alternative comparison could be between the app and web channels. This approach avoids the assumption that trends across different eye categories are similar, but it requires a separate assumption that without AR introduction, sales trends in the two online channels would be parallel.

A more robust approach that does not require either of these assumptions is the DDD approach (Angrist and Pischke 2009, p. 181; Wooldridge 2010, p.150), which combines both comparisons. Specifically, the DDD analysis measures differences between app and web sales for eyeshadows and eyeliners before and after AR introduction, relative to the same differences for other eye categories that do not have the AR feature. Thus, the DDD approach controls for both channel and category trends that could potentially confound the effect, and it relies on the more relaxed assumption that in the absence of AR introduction, sales trends in the two online channels would be parallel for products in the same category. Following Janakiraman, Lim, and Rishika (2018) and Fisher, Gallino, and Xu (2019), we conducted two falsification tests using data from the pre–AR introduction period, and the results provide support that this assumption holds in our study. Details and results for these falsification tests are included in Web Appendix F.

Accordingly, we examined changes in weekly sales for the five product categories (i.e., eyeshadow, eyeliner, eyebrows, mascara, and eye palettes) across two channels (i.e., app and web) before and after AR introduction. Our sample covers a duration of 108 weeks (i.e., 42 weeks for the pre–AR introduction period and 66 weeks for the post–AR introduction period), giving us a total of 1,080 observations ($5 \times 2 \times 108 = 1{,}080$).

*Model specification.* The outcome variable of interest was sales for category j on channel k during week t, Category $\text{Sales}_{jkt}$. As Category $\text{Sales}_{jkt}$ was a count variable with significant overdispersion (M = 64.8, SD = 78.0), we used a negative binomial model for the estimation. Following Wooldridge (2010, p.150), the vector of covariates in the regression is given by the following:

$$\begin{aligned}\mathbf{X_{jkt}}\beta = {} & \beta_0 + \beta_1 \text{AR Intro}_t + \beta_2 \text{App}_k + \beta_3 \text{AR Feature}_j \\ & + \beta_4 \text{AR Intro}_t \times \text{App}_k \times \text{AR Feature}_j \\ & + \beta_5 \text{AR Intro}_t \times \text{App}_k \\ & + \beta_6 \text{AR Intro}_t \times \text{AR Feature}_j \\ & + \beta_7 \text{App}_k \times \text{AR Feature}_j \\ & + \sum_{c=1}^{J-2} \gamma_c \text{Category}_j + \sum_{w=1}^{T-2} \delta_w \text{Week}_t + \varepsilon_{jkt}. \end{aligned} \quad (2)$$

In Equation 2, $\text{AR Intro}_t$ is a dummy variable with a value of 1 if week t was in the post–AR introduction period and 0 otherwise. $\text{App}_k$ is a dummy variable with a value of 1 for the mobile app and 0 for the website, and $\text{AR Feature}_j$ is a dummy variable with a value of 1 for eye categories with the AR feature (i.e., eyeshadow and eyeliner) and 0 for other eye categories. The key coefficient of interest was $\beta_4$, which captured the three-way interaction between AR introduction, retail channel, and categories that have the AR feature. Thus, $\beta_4$ represents the additional change in mobile app sales post–AR

**Table 5.** DDD Analysis: Impact of AR Introduction on Category Sales.

| | Column 1 Basic Pre-Post Model | Column 2 DDD Analysis | Column 3 Including Trends | Column 4 Excluding Sale Events |
|---|---|---|---|---|
| **AR Intro** | **.611 (.245) **** | .265 (.214) | .771 (.445) * | .720 (.429) * |
| App | – | .125 (.058) ** | 39.187 (9.72) *** | 31.953 (8.95) *** |
| AR Feature | – | .068 (.146) | 5.794 (8.44) | 3.720 (7.57) |
| **AR Intro × App × AR Feature** | – | **.449 (.262) *** | **.441 (.249) *** | **.465 (.264) *** |
| AR Intro × App | – | −.601 (.237) ** | .243 (.154) | .036 (.169) |
| AR Intro × AR Feature | – | −.155 (.177) | −.066 (.209) | −.113 (.200) |
| App × AR Feature | – | −.034 (.088) | −.027 (.056) | −.036 (.048) |
| Constant | 2.043 (.040) *** | 2.604 (.156) *** | 2.403 (.226) *** | 2.464 (.201) *** |
| Category dummies | Included | Included | Included | Included |
| Week dummies | Included | Included | Included | Included |
| Category trend | Not included | Not included | Included | Included |
| Channel trend | Not included | Not included | Included | Included |
| Observations | 216 | 1,080 | 1,080 | 940 |
| Log likelihood | −897 | −5,143 | −5,095 | −4,179 |

*$p \leq .10$; **$p \leq .05$; ***$p \leq .01$.
*Note*: Standard errors (clustered at category-channel level) are in parentheses.

introduction for eyeshadow and eyeliner, after accounting for channel and category-related changes over the same period (captured by $\beta_5$ and $\beta_6$ respectively). We included all lower-order interactions in the model, as well as a series of dummy variables, $Category_j$ (for $j = 1, \ldots, J$ categories) and $Week_t$, (for $t = 1, \ldots, T$ weeks) to control for category and week effects. Because $Category_j$ was perfectly collinear with AR $Feature_j$ and $Week_t$ was perfectly collinear with AR $Intro_t$, we excluded dummy variables for an additional category and week. To account for the panel nature of the data, we clustered standard errors at the category-channel level, allowing errors for observations from the same category within each channel to correlate.

*Results.* Before discussing results for the DDD analysis, we present the basic pre-post model in Table 5, Column 1. We regressed weekly mobile app sales for eyeshadow and eyeliner on AR $Intro_t$ and the vector of dummies. The coefficient for AR $Intro_t$ is significantly positive (.611, $p < .05$), providing preliminary evidence that sales increased after AR was introduced. Results for the DDD analysis are presented in Table 5, Column 2. The coefficient for the three-way interaction between AR introduction, app, and categories with the AR feature is marginally significant (.449, $p < .10$), providing some evidence that sales for eyeshadows and eyeliners increased on the app channel after AR was introduced.

*Robustness analyses.* To check the DDD identification strategy, we included channel and category trends in Equation 2 (Angrist and Pischke 2009, p. 178). Results of this alternative model are presented in Table 5, Column 3, and the coefficient of the three-way interaction of interest is similar in direction, magnitude, and significance with the main model. Although the weekly fixed effects controlled for variations in overall sales between weeks, they did not account for time-varying confounding effects that were specific to the channel-category. Thus, if there were more app-exclusive sale events for the eyeshadow and eyeliner categories in the post–AR introduction period relative to the pre–AR introduction period, the effect of AR Introduction on app sales in these two categories would be overstated. As a robustness check, we removed weeks that coincided with sale events from the analysis and present the results in Table 5, Column 4. We also split AR $Feature_j$ into the two eye categories with the AR feature, $Eyeshadow_j$ and $Eyeliner_j$, and the coefficients of both three-way interactions are marginally significant, providing convergent validity for our results. Furthermore, results from the Wald test for equality of coefficients fail to reject the null hypothesis that the coefficients are the same ($p = .76$), indicating that the effect of AR introduction on sales is not category-specific. Lastly, we estimated the same model using a Poisson regression. Results of these additional analyses are provided in Web Appendix G. Across all robustness analyses, the direction, magnitude, and significance of coefficients are similar to the main model.

Overall, the results provide additional support for $H_1$ and demonstrate that the positive impact of AR generalizes to other product categories. We note that because the retailer did not announce the introduction of AR for the eye categories, usage of the feature was low. On average, the weekly number of customers using AR to try products from the eye categories was 6.4 times lower than the number for lip categories ($M_{eyes} = 271.14$ vs. $M_{lips} = 1{,}737.00$). Thus, our result is a conservative estimate of the impact of AR introduction, and we speculate that the effect could have been larger if the retailer had advertised the feature. To establish a direct relationship between AR usage and purchase, and to further examine the moderating effects of customers' channel and category experience, we next turn our attention to the customer level.

### Customer-Level Analysis

We focused on the sample of active customers (i.e., those who made a purchase in the past one year) who browsed products in the eyeshadow or eyeliner categories during the 12-month period[2] after the retailer introduced AR for these two categories (i.e., mid-March 2018 to mid-March 2019). In total, our sample included 42,493 customers. At the time of AR introduction, 40.2% of these customers had never purchased online before (i.e., new to the online channel) and 43.4% had never purchased eyeshadow or eyeliner before (i.e., new to the categories). During the 12-month period after the retailer introduced the AR feature for the two categories, 13.9% of customers used the feature to try eyeshadows and eyeliners, and 15.0% purchased at least one product from these categories using the app. Accordingly, we modeled how AR usage influenced customers' probability of purchasing products from these two categories in the focal period.

*Model specification.* The dependent variable of interest, P(Purchase$_i^{eyes}$), was customer i's probability of purchasing at least one eyeshadow or eyeliner on the app within 12 months of AR introduction for the two categories. As the dependent variable was binary, we used a probit model with the following specification:

$$\begin{aligned} P(\text{Purchase}_i^{eyes} = 1|X_i\beta) = \Phi\,(&\beta_0 + \beta_1\,\text{AR Usage}_i^{eyes} \\ &+ \beta_2\,\text{New Channel}_i + \beta_3\,\text{New Category}_i \\ &+ \beta_4\,\text{AR Usage}_i^{eyes} \times \text{New Channel}_i \\ &+ \beta_5\,\text{AR Usage}_i^{eyes} \times \text{New Category}_i \\ &+ \mathbf{Browsing_i'}\gamma + \mathbf{PastPurchase_i'}\delta + \varepsilon_i\,). \end{aligned} \quad (3)$$

In Equation 3, Φ denotes the standard probit link function. AR Usage$_i^{eyes}$ represents the focal independent variable and takes a value of 1 if customer i used the AR feature to try eyeshadows or eyeliners during the period and 0 otherwise. New Channel$_i$ and New Category$_i$ are both indicator variables representing customers' (lack of) prior experience with the channel and category. New Channel$_i$ takes a value of 1 if customer i is new to the online channel and 0 otherwise, and New Category$_i$ takes a value of 1 if customer i is new to the two eye categories and 0 otherwise. To examine how these two variables moderate the effect of AR usage on purchase, we included interactions between the variables and AR Usage$_i^{eyes}$. We also included a vector, **Browsing$_i$**, to control for customers' browsing behavior before and during the focal period to account for customer interest and engagement. Because the browsing activity data set starts at December 2017 (i.e., three months prior to the introduction of AR for the eye categories), we used a three-month window for past browsing behavior. Lastly, we included a vector, **Past Purchase$_i$**, to control for customers' purchase history in the 12 months prior to AR introduction for the eye categories to account for customer loyalty. Table 6 provides a summary of how the variables are operationalized and their descriptive statistics. The correlations are provided in Web Appendix H, and the variance inflation factors are below 1.75, indicating that multicollinearity is not an issue.

*Identification strategy.* As customers who already intend to purchase products may be more likely to try them using the AR feature, we used the two-stage residual inclusion method to account for this self-selection bias. We used customers' past AR usage for lip products (prior to AR introduction for the eye categories) as the instrument. Customers who had used AR to try lip products in the past were already aware of the feature and could have been more likely to use it again to try eye products. Conversely, customers who had never used the feature to try lip products in the past may have been unaware of it. Because the retailer did not announce the AR introduction for the eye categories, these customers could still have been unaware of the feature. As a result, they would have been less likely to use it to try eye products. Furthermore, because lip and eye products target different areas of the face, past usage of the AR feature to try lip products should not have directly affected the probability of purchasing eye products during the focal period. Thus, we included Past AR Usage$_i^{lips}$ as an instrument in the first stage to estimate customer i's likelihood of using AR in the focal period. The variable was coded as 1 if customer i used the AR feature to try lip products in the three months before AR was introduced for the eye categories and 0 otherwise. Residuals from the first-stage estimation were then included in Equation 3 to estimate P(Purchase$_i^{eyes}$). Similar to the product model, we bootstrapped 1,000 samples to obtain the proper standard errors. To examine if the findings are robust to alternative identification strategies, we also adopted the propensity score weighting approach, which does not rely on instruments. We discuss this further in the robustness analyses section.

*Results.* The coefficient of the instrument in the first stage estimation (provided in Web Appendix I) is positive and significant (.176, $p < .01$), and the F-statistic of excluded instrument in the first stage regression is 15.8, providing evidence for the strength of the instrument. However, the coefficient for the residual correction term, which is equivalent to the Hausman test for the presence of endogeneity (Papies, Ebbes, and Van Heerde 2017), is not significant, suggesting that endogeneity may not be a concern. We also used the Heckman selection method (Heckman 1979) as an alternative identification strategy, and the inverse Mills ratio is similarly not significant. Therefore, we report estimates for the uncorrected model in the results section and provide the full result for both the two-stage residual inclusion and Heckman selection methods in Web Appendix I. We note that across all models, the substantive findings of interest remain consistent.

Table 7, Column 1, displays the results for the model without interactions, representing factors influencing the purchase of eyeshadows or eyeliners during the 12 months after AR introduction for these categories. The coefficient of AR

[2] We also repeated the analysis for the three- and six-month periods and discuss insights from these analyses in the "Results" section.

**Table 6.** Variable Operationalization and Descriptive Statistics for Customer Model.

| Variable | Operationalization | Mean | SD | Min | Median | Max |
|---|---|---|---|---|---|---|
| Purchase$^{eyes}$ (1/0) | 1 if customer bought eye products in the focal period, 0 otherwise | .15 | .36 | .00 | .00 | 1.00 |
| AR Usage$^{eyes}$ (1/0) | 1 if AR was used to try eye products in the focal period, 0 otherwise | .14 | .35 | .00 | .00 | 1.00 |
| New Channel (1/0) | 1 if customer had never purchased from the retailer's online channel before the focal period, 0 otherwise | .40 | .49 | .00 | .00 | 1.00 |
| New Category (1/0) | 1 if customer had never purchased eye products from the retailer before the focal period, 0 otherwise | .43 | .50 | .00 | .00 | 1.00 |
| **Browsing controls** | | | | | | |
| Past Duration | Total browsing duration in the past three months (in minutes) | 11.56 | 25.60 | .00 | .00 | 150.18 |
| Past Pages$^{eyes}$ | Number of eye product pages viewed in the past three months | .77 | 2.90 | .00 | .00 | 23.00 |
| Past AR Usage$^{lips}$ (1/0) | 1 if customer had used AR for lip products in the past three months, 0 otherwise | .02 | .15 | .00 | .00 | 1.00 |
| Duration | Total browsing duration in the focal period (in minutes) | 153.17 | 148.32 | 2.82 | 106.57 | 1054.25 |
| Pages$^{eyes}$ | Number of eye product pages viewed in the focal period | 19.21 | 24.98 | 1.00 | 9.00 | 128.00 |
| **Purchase controls** | | | | | | |
| Past Order Number | Number of transactions in the past one year | 6.92 | 5.83 | 1.00 | 5.00 | 37.00 |
| Past Order Value | Average value of transactions in the past one year | 81.88 | 50.49 | .00 | 70.68 | 547.00 |
| Past Eye Purchases | Number of eye products purchased in the past one year | .89 | 1.50 | .00 | .00 | 11.00 |
| Recent Order | Number of months from the most recent transaction | 2.64 | 2.49 | .03 | 2.03 | 12.13 |
| First Order | Number of months from the first transaction | 18.92 | 7.73 | .03 | 22.37 | 26.80 |

*Notes*: "Eye products" refers to eyeshadow and eyeliner, the two categories of interest. The focal period is 12 months after the retailer introduced AR for eye products (i.e., March 15, 2018, to March 15, 2019).

Usage$^{eyes}$ is positive and significant (.046, $p < .05$), providing further evidence for $H_1$. The coefficients of other variables are largely in line with expectations. For example, the coefficient for New Channel ($-.329$, $p < .01$) and New Category ($-.120$, $p < .01$) are significantly negative, indicating that customers who are new to the online channel or product category are less likely to make a purchase. The number of orders (.007, $p < .01$), average order value (.002, $p < .01$), and number of eye products purchased in the past (.080, $p < .01$) are positively related to probability of purchasing eye products. Furthermore, total browsing duration (.000, $p < .01$) and number of eye product pages viewed (.007, $p < .01$) are also positively related to the purchase of eye products.

Table 7, Column 2, provides results for the 12-month model, including interactions. The interaction between AR Usage$^{eyes}$ and New Channel is positive and significant (.091, $p < .05$), suggesting that AR has a stronger effect among customers who had never purchased online in the past. The average marginal effect of AR usage for customers who are new to the online channel is significantly positive (.018, $p < .01$), but this effect is not significant for existing online customers (.004, $p = .59$). Thus, $H_{3a}$ is supported. While the interaction between AR Usage$^{eyes}$ and New Category is marginally significant (.082, $p < .10$), the average marginal effect of AR usage is significantly positive for customers who are new to the product category (.019, $p < .01$) and not significant for existing category customers (.003, $p = .65$), providing support for $H_{3b}$ as well.

To understand how the impact of AR changes over time, we repeated the same analysis using a six-month and three-month window, presented in Columns 3 and 4 of Table 7. We find that the interactions between AR Usage$^{eyes}$ and both New Channel and New Category become stronger over time. Although both interactions (as well as the average marginal effects) are insignificant in the 3-month period, the interaction with New Channel becomes significant in the 6- and 12-month period, and the interaction with New Category becomes marginally significant in the 12-month period. Similar to the 12-month model, the average marginal effects in the 6-month model are significantly positive for customers who are new to the online channel (.022, $p < .01$ vs. .006, $p = .44$ for existing online customers) and product category (.019, $p < .05$ vs. .007, $p = .31$ for existing category customers). These results suggest that customers may require some time to become comfortable with the technology before using it to make purchase decisions. In addition, the results also imply that the impact of AR does not wear out over time, which rules out novelty effects as an alternative explanation.

*Robustness analyses.* Results for the two-stage residual inclusion and Heckman selection methods for all 3-, 6-, and 12-month periods are provided in Web Appendix I. To further examine if the findings are robust to alternative identification strategies, we applied the propensity score weighting approach. We used the first stage equation to calculate customers' propensity for using AR in the focal period and include this as weights in the estimation of Equation 3, following Bell, Gallino, and Moreno (2018). The results are consistent with the main model and are also reported in Web Appendix I.

We also examined if the findings are robust to alternative variable operationalizations. First, instead of the probability of purchasing eye products, we used the number of eye products purchased during the focal period as an alternative dependent variable. Second, we replaced the binary AR Usage variable

**Table 7.** Customer Model: Impact of AR Usage on Probability of Purchase and Moderating Effects of Customer Characteristics.

| | Column 1 12-Month Model (Without Interactions) | Column 2 12-Month Model (Full Model) | Column 3 6-Month Model (Full Model) | Column 4 3-Month Model (Full Model) |
|---|---|---|---|---|
| **AR Usage$^{eyes}$ ($H_1$)** | **.046 (.022) **** | −.015 (.030) | .004 (.047) | −.036 (.084) |
| New Channel | −.329 (.018) *** | −.344 (.019) *** | −.323 (.029) *** | −.296 (.040) *** |
| New Category | −.121 (.020) *** | −.134 (.021) *** | −.060 (.032) * | −.122 (.042) *** |
| **AR Usage$^{eyes}$ × New Channel ($H_{3a}$)** | **–** | **.091 (.046) **** | **.144 (.070) **** | **.091 (.138)** |
| **AR Usage$^{eyes}$ × New Category ($H_{3b}$)** | **–** | **.082 (.045) *** | **.075 (.068)** | **−.004 (.135)** |
| Past Duration | −.002 (.000) *** | −.002 (.000) *** | −.002 (.000) *** | −.002 (.000) *** |
| Past Pages$^{eyes}$ | −.003 (.003) | −.003 (.003) | .000 (.002) | −.004 (.002) ** |
| Duration | .000 (.000) *** | .000 (.000) *** | .001 (.000) *** | .001 (.000) *** |
| Pages$^{eyes}$ | .007 (.000) *** | .007 (.000) *** | .008 (.000) *** | .017 (.001) *** |
| Past Order Number | .007 (.002) *** | .007 (.002) *** | .005 (.003) | .003 (.004) |
| Past Order Value | .002 (.000) *** | .002 (.000) *** | .002 (.000) *** | .002 (.000) *** |
| Past Eye Purchases | .080 (.006) *** | .081 (.006) *** | .103 (.012) *** | .062 (.011) *** |
| Recent Order | −.019 (.004) *** | −.020 (.004) *** | −.020 (.006) *** | −.016 (.008) * |
| First Order | −.001 (.001) | −.001 (.001) | −.002 (.002) | .005 (.002) * |
| Constant | −1.285 (.035) *** | −1.276 (.035) *** | −1.567 (.054) *** | −1.771 (.077) *** |
| Observations | 42,493 | 42,493 | 24,147 | 13,434 |
| Log likelihood | −16,614 | −16,609 | −7,423 | −3,773 |

*$p \leq .10$; **$p \leq .05$; ***$p \leq .01$
*Note*: Standard errors are in parentheses.

with the number of sessions involving AR usage during the focal period. Third, as alternative measures of channel and category experience, we used the number of online transactions and number of eye products purchased prior to AR introduction for the eye categories, respectively. Findings from these models are consistent, and the results are presented in Web Appendix J.

## Discussion

Although firms are keen to invest in AR, research demonstrating its impact in real-world contexts is limited. The present research provides some preliminary confirmation that both the availability and usage of AR have a small but positive impact on sales. Taken together, our findings provide converging evidence that AR is most effective when product-related uncertainty is high, indicating that uncertainty reduction could be a possible mechanism through which AR could improve sales. Nevertheless, we do not find a significant moderating effect for product ratings, suggesting that even though AR may reduce product fit uncertainty, it may still be unable to compensate for the higher performance uncertainty associated with products that have lower ratings.[3] Although we have adopted instrumental variable and quasi-experimental approaches to address endogeneity that is inherent in observational data, we acknowledge that these findings should be viewed as evidence based on correlations, with attempts to come close to causality.

### Research Implications

*Augmented reality and product preference.* Complementing past research that has explored how website features drive sales for niche products (e.g., Brynjolfsson, Hu, and Simester 2011; Tucker and Zhang 2011), we show that AR can increase preference for products or brands that are less popular. Thus, retailers carrying wide product assortments can use AR to stimulate demand for products in the long tail of the sales distribution. AR may also help level the playing field for less popular brands. With the launch of AR-enabled display ads on advertising platforms such as Facebook and YouTube, less-established brands could consider investing in this new ad format, as they stand to benefit most from this technology. Retailers selling premium products may also leverage AR to improve decision comfort and reduce customers' hesitation in the purchase process.

*Augmented reality and category sales.* We find that the impact of AR is stronger for customers who are new to the product category, suggesting that AR could increase sales via category expansion. However, because AR seems to be most effective when the level of uncertainty is high, its impact may diminish over time as customers become more familiar with the product category and experience less uncertainty.[4] Nevertheless, the finding that AR has a stronger impact for products that are more expensive suggests that, beyond increasing unit sales, AR can also improve category revenues by encouraging

[3] We thank an anonymous reviewer for suggesting this possible explanation for the lack of a significant result that supports $H_{2c}$.

[4] We thank an anonymous reviewer for highlighting this possibility, and we encourage future research to explore the dynamic effects of AR usage.

customers to purchase products with wider margins. Thus, investments in deploying AR in retail could pay off in the long run.

*Augmented reality and channel choice.* Compared with customers who are already familiar with purchasing online, we find that AR has a stronger effect for customers who are new to the online channel. As prior research has shown that multichannel customers are more profitable (Montaguti, Neslin, and Valentini 2016), omnichannel retailers can use AR to encourage their offline customers to adopt the online channel. Given that AR increases online sales among customers who are new to the channel, a potential concern is that AR could lead to cannibalization of sales from offline channels. To understand if the increase in app purchases that we observed was happening at the expense of other sales channels, we ran the same model in Equation 3 but replaced the dependent variable with the probability of purchasing eye products in the web and offline channels (results reported in Web Appendix K). We did not find evidence to indicate that offline customers who use AR (on the app) are more likely to purchase from the web, suggesting that the impact of AR is specific to the app platform. Interestingly, we find that offline customers who use AR are more likely to purchase from the offline channel in the three-month model but not in the six and 12-month model. Thus, contrary to our expectations, the results suggest that AR could have a positive spillover effect to the offline channel, at least in the short run.

## An Agenda for Future Research

Complementing prior research, which has predominantly studied AR from a consumer perspective, our research extends the literature by examining what AR means for retailers. To encourage the academic community to produce more impactful research in this nascent field, we developed a research agenda for AR in marketing, with an emphasis on identifying research topics that have strong managerial relevance for industry practitioners. Drawing on a review of the academic literature (e.g., Wedel, Bigné, and Zhang 2020) and recent advancements in AR technology, we generated a list of potential research topics and synthesized these topics into five themes. Next, we consulted two senior marketing practitioners and two academics with expertise in this area to review the research themes and associated topics, and we refined the list according to their feedback.

To determine the practical importance of each research theme, we conducted an online survey with 36 marketing practitioners from companies that were using (or planning to use) AR in their marketing, advertising, or retailing activities. Survey respondents first independently rated each research theme in terms of importance to business performance (see Stremersch and Van Dyck 2009) before ranking the five research themes from most to least important. To avoid primacy and recency effects, the order of research themes was randomized across respondents. The mean rating (ranging from 5.1 to 5.8 on a seven-point scale) and ranking scores (from 1 to 5; lower number reflects higher importance) are inversely proportional, demonstrating internal consistency. Web Appendix L provides details for the survey, including survey design, respondent recruitment, and background of respondents.

Table 8 presents the research agenda for AR in marketing, comprising the five research themes (ordered by practical importance) and potential topics that could be explored under each theme. Given the novelty of the technology, marketers were primarily concerned with how different design features could be configured to create more effective AR experiences for consumers. For example, greater clarity is needed regarding factors that affect AR experiences, such as fidelity (i.e., how closely virtual objects resemble real objects), motion (i.e., static vs. animated virtual objects), spatial presence (i.e., the feeling that virtual objects exist in a physical space), and embodiment (i.e., the ability to use bodily movements to control virtual objects), and how these can be delivered on AR interfaces. Beyond visual and auditory senses, how haptic feedback (e.g., emission of vibrations on devices to stimulate the sense of touch) influences AR experiences is also of interest.

Another important consideration is how AR fits into companies' overall marketing strategy. Specifically, marketers would like to know how they can better integrate AR at different stages of the customer journey to increase brand engagement, build emotional connections, and improve relationships with customers. There is also ambiguity regarding the synergy between AR and other elements of the marketing communications mix (e.g., advertising, sales promotions), as well as the effectiveness of product placements and pop-up stores in AR-enabled virtual environments. In particular, the potential for this new technology to complement or replace existing communication and retail channels is still uncertain. As most recent applications of AR have focused on consumer products, marketers also need more guidance on how AR can be appropriately deployed in service industries, such as the tourism and hospitality sector.

Besides these two key areas, other worthwhile avenues to explore include the impact of AR on consumer behavior (e.g., cognitive functions, rational decision making, and brand perceptions), how marketers can promote wider adoption of AR, and how the technology can be used to generate valuable marketing insights. Although our research agenda focuses on AR, we note that the research themes could be broadened to encompass other extended realities (i.e., virtual reality and mixed reality).

In conclusion, we believe that the marketing community would benefit from a deeper investigation of virtual experiences and their role in marketing. We are excited about where this field is heading, and we look forward to more insightful research to reinforce our understanding of the profound impacts of these new technologies in the marketing domain.

**Table 8.** Research Agenda for AR in Marketing.

| Research Themes | Potential Research Topics |
|---|---|
| Designing effective AR experiences<br>Rating: 5.81<br>Ranking: 2.14 | • How factors such as fidelity (i.e., how closely virtual objects resemble real objects), motion (i.e., static vs. animated virtual objects), spatial presence (i.e., the feeling that virtual objects exist in a physical space), and embodiment (i.e., the ability to use bodily movements to control virtual objects) affect AR experiences.<br>• How the incorporation of senses such as haptic feedback (e.g., emission of vibrations) influences AR experiences.<br>• How content and other elements in the virtual environment could be personalized to enhance AR experiences and influence behavior. |
| AR and marketing strategy<br>Rating: 5.50<br>Ranking: 2.78 | • How marketers could use AR more effectively at different stages of the customer journey to increase brand engagement and improve relationships with customers.<br>• Synergy between AR and other elements in the marketing communications mix (e.g., advertising, sales promotions).<br>• Effectiveness of product placements and pop-up stores in AR-enabled virtual environments, as well as their potential to complement or replace physical stores.<br>• How AR could be deployed in service industries (e.g., tourism and hospitality, food and beverage retail). |
| AR and consumer behavior<br>Rating: 5.22<br>Ranking: 3.06 | • How AR experiences affect sensory perceptions and cognitive functions (e.g., attention, information processing, learning, and memory).<br>• How AR experiences affect rational decision making (e.g., product selection strategies, relative importance of attributes) and irrational tendencies (e.g., psychological ownership).<br>• The role of AR experiences in attitude formation and brand perceptions.<br>• How AR experiences affect purchase behaviors and postpurchase product evaluations. |
| Promoting AR adoption<br>Rating: 5.22<br>Ranking: 3.42 | • Exploring barriers to consumers' use of AR technologies (e.g., awkwardness of using it in public, privacy and security concerns, lack of realism in virtual environments) and how marketers can overcome these barriers to encourage wider adoption.<br>• How delivery of the AR experience and advancements in high-tech devices (e.g., 3D depth camera technology, wearable AR glasses) influence consumers' acceptance and usage of the technology.<br>• How offline contextual factors (e.g., distance to physical stores, private vs. public space) affect AR usage. |
| AR as a marketing intelligence tool<br>Rating: 5.11<br>Ranking: 3.61 | • How AR experiences could be used to generate insights for new product development, assortment planning, and store layout/design.<br>• Identifying new behavioral data (e.g., motion, interactions within virtual environments) that could be obtained from AR platforms, as well as how such data could be used to measure/predict consumers' responses or decision-making processes.<br>• Privacy and security concerns regarding behavioral data collected on AR platforms and what marketers can do to reduce these concerns. |

*Notes*: Research themes are ordered by importance on the basis of surveys with 36 marketing practitioners. Details of the survey are provided in Web Appendix L. "Rating" refers to the mean importance rating score (on a seven-point scale). "Ranking" refers to the mean importance ranking (from 1 to 5; lower number reflects higher importance).

## Acknowledgments

This research is based on part of the first author's dissertation at Singapore Management University and was supported by the LVMH-SMU Luxury Brand Research Initiative and the SMU Retail Centre of Excellence. The authors thank Ernst Christiaan Osinga, Kapil R. Tuli, and Sunil Gupta for their valuable feedback on an earlier version of this article; Vanitha Swaminathan, Simon Torring, Tito Costa, and Christian Sandor for their feedback on the research agenda; and industry practitioners who participated in the survey. The authors also gratefully acknowledge the *JM* review team for their thoughtful comments and helpful suggestions throughout the review process.

## Associate Editor

Benedict Dellaert

## Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

## Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This research received financial support from the Singapore Economic Development Board (Grant Number: EDB-EDAS-2019-1).

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## References


Angrist, Joshua D. and Jörn-Steffen Pischke (2009), *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton, NJ: Princeton University Press.

Arora, Sandeep, Frenkel ter Hofstede, and Vijay Mahajan (2017), "The Implications of Offering Free Versions for the Performance of Paid Mobile Apps," *Journal of Marketing*, 81 (6), 62–78.

Arthur, Rachel (2017), "Augmented Reality Is Set to Transform Fashion and Retail," *Forbes* (October 31), https://www.forbes.com/sites/rachelarthur/2017/10/31/augmented-reality-is-set-to-transform-fashion-and-retail/.

Bauer, Raymond A. (1960), "Consumer Behavior as Risk Taking," in *Dynamic Marketing for a Changing World*, Robert S. Hancock, ed. Chicago: American Marketing Association, 389–98.

BCG (2018), "Augmented Reality: Is the Camera the Next Big Thing in Advertising?" (April 3), https://www.bcg.com/publications/2018/augmented-reality-is-camera-next-big-thing-advertising.

Bell, David R., Santiago Gallino, and Antonio Moreno (2018), "Offline Showrooms in Omnichannel Retail: Demand and Operational Benefits," *Management Science*, 64 (4), 1629–51.

Brengman, Malaika, Kim Willems, and Helena van Kerrebroeck (2019), "Can't Touch This: The Impact of Augmented Reality Versus Touch and Non-Touch Interfaces on Perceived Ownership," *Virtual Reality*, 23 (3), 269–80.

Brynjolfsson, Erik, Hu Yu (Jeffrey), and Duncan Simester (2011), "Goodbye Pareto Principle, Hello Long Tail: The Effect of Search Costs on the Concentration of Product Sales," *Management Science*, 57 (8), 1373–86.

Burke, Raymond R. (2002), "Technology and the Customer Interface: What Consumers Want in the Physical and Virtual Store," *Journal of the Academy of Marketing Science*, 30 (4), 411–32.

Cameron, A. Colin and Douglas L. Miller (2015), "A Practitioner's Guide to Cluster-Robust Inference," *Journal of Human Resources*, 50 (2), 317–72.

Carnahan, Daniel (2019), "Facebook Has Made AR Ads Available to All Marketers Through Its Ad Manager," *Business Insider* (December 12), https://www.businessinsider.com/facebook-experiments-with-augmented-reality-advertising-2019-12.

Chandukala, Sandeep R., Jeffrey P. Dotson, and Qing Liu (2017), "An Assessment of When, Where, and Under What Conditions In-Store Sampling Is Most Effective," *Journal of Retailing*, 93 (4), 493–506.

Chaykowski, Kathleen (2018), "Inside Facebook's Bet on an Augmented Reality Future," *Forbes* (March 8), https://www.forbes.com/sites/kathleenchaykowski/2018/03/08/inside-facebooks-bet-on-an-augmented-reality-future/#6973829b4d56.

Chen, Yubo and Jinhong Xie (2008), "Online Consumer Review: Word-of-Mouth as a New Element of Marketing Communication Mix," *Management Science*, 54 (3), 477–91.

Danaher, Peter J., Tracey S. Danaher, Michael Stanley Smith, and Ruben Loaiza-Maya (2020), "Advertising Effectiveness for Multiple Retailer-Brands in a Multimedia and Multichannel Environment," *Journal of Marketing Research*, 57 (3), 445–67.

Danaher, Peter J., Isaac W. Wilson, and Robert A. Davis (2003), "A Comparison of Online and Offline Consumer Brand Loyalty," *Marketing Science*, 22 (4), 461–76.

Dimoka, Angelika, Yili Hong, and Paul A. Pavlou (2012), "On Product Uncertainty in Online Markets: Theory and Evidence," *MIS Quarterly*, 36 (2), 395–426.

Erdem, Tülin, Joffre Swait, and Ana Valenzuela (2006), "Brands as Signals: A Cross-Country Validation Study," *Journal of Marketing*, 70 (1), 34–49.

Ericsson (2017), "Merged Reality: Understanding How Virtual and Augmented Realities Could Transform Everyday Reality," (accessed June 26, 2020), https://www.ericsson.com/en/reports-and-papers/consumerlab/reports/merged-reality.

Fisher, Marshall L., Santiago Gallino, and Xu Joseph Jiaqi (2019), "The Value of Rapid Delivery in Omnichannel Retailing," *Journal of Marketing Research*, 56 (5), 732–48.

Gallino, Santiago and Antonio Moreno (2018), "The Value of Fit Information in Online Retail: Evidence from a Randomized Field Experiment," *Manufacturing & Service Operations Management*, 20 (4), 767–87.

Gray, Alistair (2019), "Retailers Grapple with $100bn Returns Problem," *Financial Times* (December 27), https://www.ft.com/content/5bafd9c0-235f-11ea-92da-f0c92e957a96.

Griffin, Andrew (2017), "Apple's Tim Cook on iPhones, Augmented Reality, and How He Plans to Change Your World," *The Independent* (October 12), https://www.independent.co.uk/life-style/gadgets-and-tech/features/apple-iphone-tim-cook-interview-features-new-augmented-reality-ar-arkit-a7993566.html.

Gu, Zheyin (Jane) and Giri K. Tayi (2015), "Consumer Mending and Online Retailer Fit-Uncertainty Mitigating Strategies," *Quantitative Marketing and Economics*, 13 (3), 251–82.

Heckman, James J. (1979), "Sample Selection Bias as a Specification Error," *Econometrica*, 47 (1), 153–62.

Heller, Jonas, Mathew Chylinski, Ko de Ruyter, Dominik Mahr, and Debbie I. Keeling (2019a), "Let Me Imagine That for You: Transforming the Retail Frontline Through Augmenting Customer Mental Imagery Ability," *Journal of Retailing*, 95 (2), 94–114.

Heller, Jonas, Mathew Chylinski, Ko de Ruyter, Dominik Mahr, and Debbie I. Keeling (2019b), "Touching the Untouchable: Exploring Multi-Sensory Augmented Reality in the Context of Online Retailing," *Journal of Retailing*, 95 (4), 219–34.

Hilken, Tim, Debbie I. Keeling, Ko de Ruyter, Dominik Mahr, and Mathew Chylinski (2020), "Seeing Eye to Eye: Social Augmented Reality and Shared Decision Making in the Marketplace," *Journal of the Academy of Marketing Science*, 48 (2), 143–64.

Hilken, Tim, Ko de Ruyter, Mathew Chylinski, Dominik Mahr, and Debbie I. Keeling (2017), "Augmenting the Eye of the Beholder: Exploring the Strategic Potential of Augmented Reality to Enhance Online Service Experiences," *Journal of the Academy of Marketing Science*, 45 (6), 884–905.

Hoch, Stephen J. (2002) "Product Experience Is Seductive," *Journal of Consumer Research*, 29 (3), 448–54.

Hollenbeck, Brett (2018), "Online Reputation Mechanisms and the Decreasing Value of Chain Affiliation," *Journal of Marketing Research*, 55 (5), 636–54.

Hong, Yili and Paul A. Pavlou (2014), "Product Fit Uncertainty in Online Markets: Nature, Effects, and Antecedents," *Information Systems Research*, 25 (2), 328–44.

Janakiraman, Ramkumar, Joon Ho Lim, and Rishika Rishika (2018), "The Effect of a Data Breach Announcement on Customer Behavior: Evidence from a Multichannel Retailer," *Journal of Marketing*, 82 (2), 85–105.

Kempf, Deanna S. and Robert E. Smith (1998), "Consumer Processing of Product Trial and the Influence of Prior Advertising: A Structural Modeling Approach," *Journal of Marketing Research*, 35 (3), 325–39.

Kim, Jiyeon and Sandra Forsythe (2008), "Adoption of Virtual Try-On Technology for Online Apparel Shopping," *Journal of Interactive Marketing*, 22 (2), 45–59.

Kim, Youngsoo and Ramayya Krishnan (2015), "On Product-Level Uncertainty and Online Purchase Behavior: An Empirical Analysis," *Management Science*, 61 (10), 2449–67.

Kübler, Raoul, Koen Pauwels, Gökhan Yildirim, and Thomas Fandrich (2018), "App Popularity: Where in the World Are Consumers Most Sensitive to Price and User Ratings?" *Journal of Marketing*, 82 (5), 20–44.

Metz, Rachel (2019), "Virtual Makeovers Are Better than Ever. Beauty Companies Are Trying to Cash In," CNN Business (February 19), https://edition.cnn.com/2019/02/19/tech/augmented-reality-makeup/index.html.

Montaguti, Elisa, Scott A. Neslin, and Sara Valentini (2016), "Can Marketing Campaigns Induce Multichannel Buying and More Profitable Customers? A Field Experiment," *Marketing Science*, 35 (2), 201–17.

Papies, Dominik, Peter Ebbes, and Harald J. van Heerde (2017), "Addressing Endogeneity in Marketing Models," in *Advanced Methods for Modeling Markets*, Peter S. H. Leeflang, Jaap E. Wieringa, Tammo H. A. Bijmolt and Koen H. Pauwels, eds. Cham, Switzerland: Springer, 581–627.

Pavlou, Paul. A., Huigang Liang, and Yajiong Xue (2007), "Understanding and Mitigating Uncertainty in Online Exchange Relationships: A Principal-Agent Perspective," *MIS Quarterly*, 31 (1), 105–36.

Petrin, Amil and Kenneth Train (2010), "A Control Function Approach to Endogeneity in Consumer Choice Models," *Journal of Marketing Research*, 47 (1), 3–13.

Petrock, Victoria (2020), "US Virtual and Augmented Reality Users 2020," eMarketer (April 7), https://www.emarketer.com/content/us-virtual-and-augmented-reality-users-2020.

Porter, Michael E. and James E. Heppelmann (2017), "Why Every Organization Needs an Augmented Reality Strategy," *Harvard Business Review*, 95 (6), 46–57.

Simonson, Itamar (1992), "The Influence of Anticipating Regret and Responsibility on Purchase Decisions," *Journal of Consumer Research*, 19 (1), 105–18.

Smith, Robert E. and William R. Swinyard (1982), "Information Response Models: An Integrated Approach," *Journal of Marketing*, 46 (1), 81–93.

Stremersch, Stefan and Walter van Dyck (2009), "Marketing of the Life Sciences: A New Framework and Research Agenda for a Nascent Field," *Journal of Marketing*, 73 (4), 4–30.

Terza, Joseph V., Anirban Basu, and Paul J. Rathouz (2008), "Two-Stage Residual Inclusion Estimation: Addressing Endogeneity in Health Econometric Modeling," *Journal of Health Economics*, 27 (3), 531–43.

Tucker, Catherine (2014), "Social Networks, Personalized Advertising, and Privacy Controls," *Journal of Marketing Research*, 51 (5), 546–62.

Tucker, Catherine and Juanjuan Zhang (2011), "How Does Popularity Information Affect Choices? A Field Experiment," *Management Science*, 57 (5), 828–42.

Weathers, Danny, Subhash Sharma, and Stacy L. Wood (2007), "Effects of Online Communication Practices on Consumer Perceptions of Performance Uncertainty for Search and Experience Goods," *Journal of Retailing*, 83 (4), 393–401.

Wedel, Michel, Enrique Bigné, and Jie Zhang (2020), "Virtual and Augmented Reality: Advancing Research in Consumer Marketing," *International Journal of Research in Marketing*, 37 (3), 443–465.

Wood, Stacy L. (2001), "Remote Purchase Environments: The Influence of Return Policy Leniency on Two-Stage Decision Processes," *Journal of Marketing Research*, 38 (2), 157–69.

Wooldridge, Jeffrey M. (2010), *Econometric Analysis of Cross Section and Panel Data*, 2nd ed. Cambridge, MA: MIT Press.

Yim, Mark Yi-Cheon, Shu-Chuan Chu, and Paul L. Sauer (2017), "Is Augmented Reality Technology an Effective Tool for E-Commerce? An Interactivity and Vividness Perspective," *Journal of Interactive Marketing*, 39, 89–103.